Algebraic Vocabulary

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Algebraic vocabulary

What is a term?

- A **term** is either:
  - a letter (**variable**) on its own, or a variable raised to a power
    - For example, x or x^2
  - a number on its own
    - For example, 20
    - These are also called **constants** as they can't change value
  - or a number **multiplied** by a letter
    - For example, $5x$
- The number in front of a letter is called a **coefficient**
  - The coefficient of x in $6x$ is 6
  - The coefficient of y in $-5y$ is -5
- Terms can include **powers** and more than one letter,
  - $6xy$
  - $4x^2$
  - ab^3c

What is a factor?

- A **factor** is any number or letter that **divides** a term **exactly**
  - There is no remainder
    - The factors of $3x$ are 1, 3, x and $3x$
    - The factors of $4xy$ are 1, 2, 4, x, $2x$, $4x$, y, $2y$, $4y$, xy, $2xy$ and $4xy$
- A term can be separated into **factors** that **multiply** together to give that term
  - Two factors of $5x$ are 5 and x
    - $5 \times x = 5x$
- To **factorise** means to write something as a **multiplication** of **factors**
- When comparing two terms, a **common factor** is one that divides both
  - Compare $6xy$ with $4x$
    - Common factors are 1, 2, x and $2x$
    - The **highest** (or greatest) common factor is $2x$

What is an expression?

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

- An expression is made by adding, subtracting, multiplying or dividing **terms**
  - $2x + 5y$
  - $b^2 - 2cd$
  - $\frac{6y}{5t}$
  - A single term is still an expression

- Expressions can be **simplified** (made easier)
  - $x + x + x$ simplifies to $3x$

What is an equation?

- An **equation** is an algebraic statement with an **equals sign** between a **left-hand side** and a **right-hand side**
  - Both sides are **equal** in value
  - For example, if $2x$ has the same value as 10, then $2x = 10$

- An equation can be **solved** by finding the values of the letters that make both sides equal
  - The equation $2x = 10$ is solved when x has the value of 5
    - $x = 5$ is called the **solution**

What is a formula?

- A **formula** is a rule, definition or relationship between different quantities, written in shorthand using **letters**
  - For example, weight, w, is mass, m, multiplied by gravitational acceleration, g
    - The formula is $w = mg$

- It is common to **substitute** numbers into a formula
  - But a formula on its own cannot be solved

- To turn a **formula** into an **equation**, more information is needed
  - For example, if $w = 50$ and $m = 5$, the formula $w = mg$ becomes the equation $50 = 5g$

Examiner Tips and Tricks

- Knowing the differences between an **expression**, an **equation** and a **formula** will help you to understand the wording of exam questions.

Worked Example

(a) From the list below, write down

(i) an expression,

Answer:

(i)

An expression does not have an equals sign

$7x - 9$ is the expression

(ii)

An equation has an equals sign and can be solved

$2x + 5 = 4$ is the equation

(b) If $x = 10$, $v = 2$ and $w = 3$, use the formula shown to write an equation in t.

Answer:

$x = vt - w$ is the formula shown (a group of different quantities forming a relationship)

Substitute $x = 10$, $v = 2$ and $w = 3$ into the formula

$10 = 2t - 3$

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